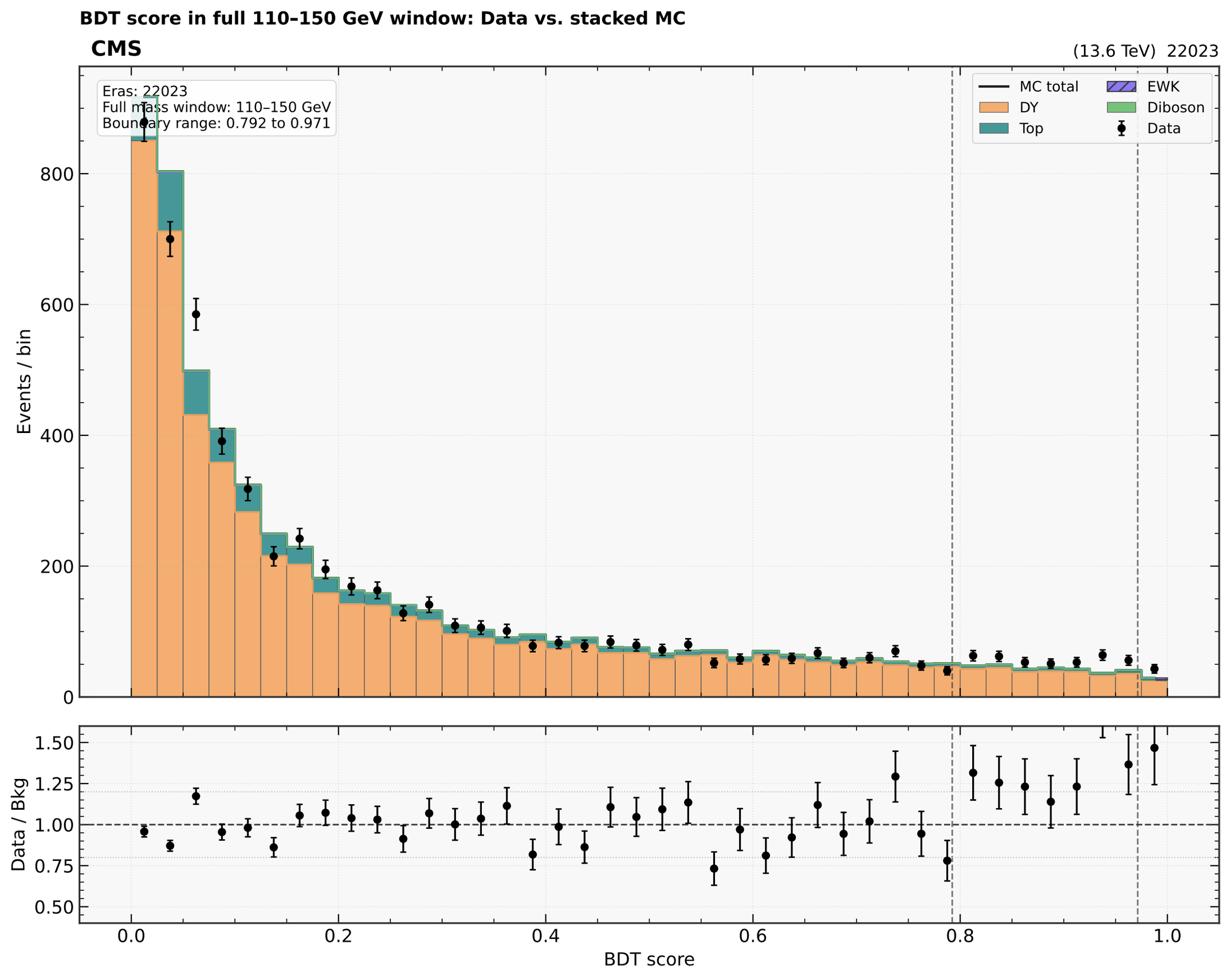
Run-3 H → μμ VBF BDT Category Optimisation

Eras : 22023
Category edges : 0.0000, 0.7923, 0.9365, 0.9714, 1.0000
MC scale factor : 1.0000 (Data/MC sideband normalisation)
Blinding window : 115–135 GeV
Sidebands : (110,115) ∪ (135,150) GeV
Outputs dir : /eos/home-m/memmad/HiggstoDimuon_VBF/src/drivers/9_TMVA/Outputs
Extra selection : –

CONTENTS:

p.2 BDT score distribution: Data vs. MC
p.3 BDT bin yields
p.4 Significance vs. number of categories
p.5 Boundary scan step-by-step (AN Fig. 32)
p.6 Atanh BDT + S/√B per category
p.7 BDT output vs. mass hypothesis (AN Fig. 26R)
p.8 Signal FWHM vs. BDT score
p.9 ROC curve with AUC
p.10 Blinded dimuon mass spectrum
p.11 Mass sculpting check (AN Fig. 25)
p.12 Per-category mass distributions and background fits
p.13 Background model comparison (χ²/ndf)
p.14 VBF signal DCB fits per category
p.15 DCB parameter table
p.16 σ_CB vs. category
p.17 Signal acceptance × efficiency
p.18 Per-category yield and sensitivity summary
p.19 Signal mass shapes mH=120/125/130 (AN Fig. 40)
p.20 Category summary table
p.21 Category process composition

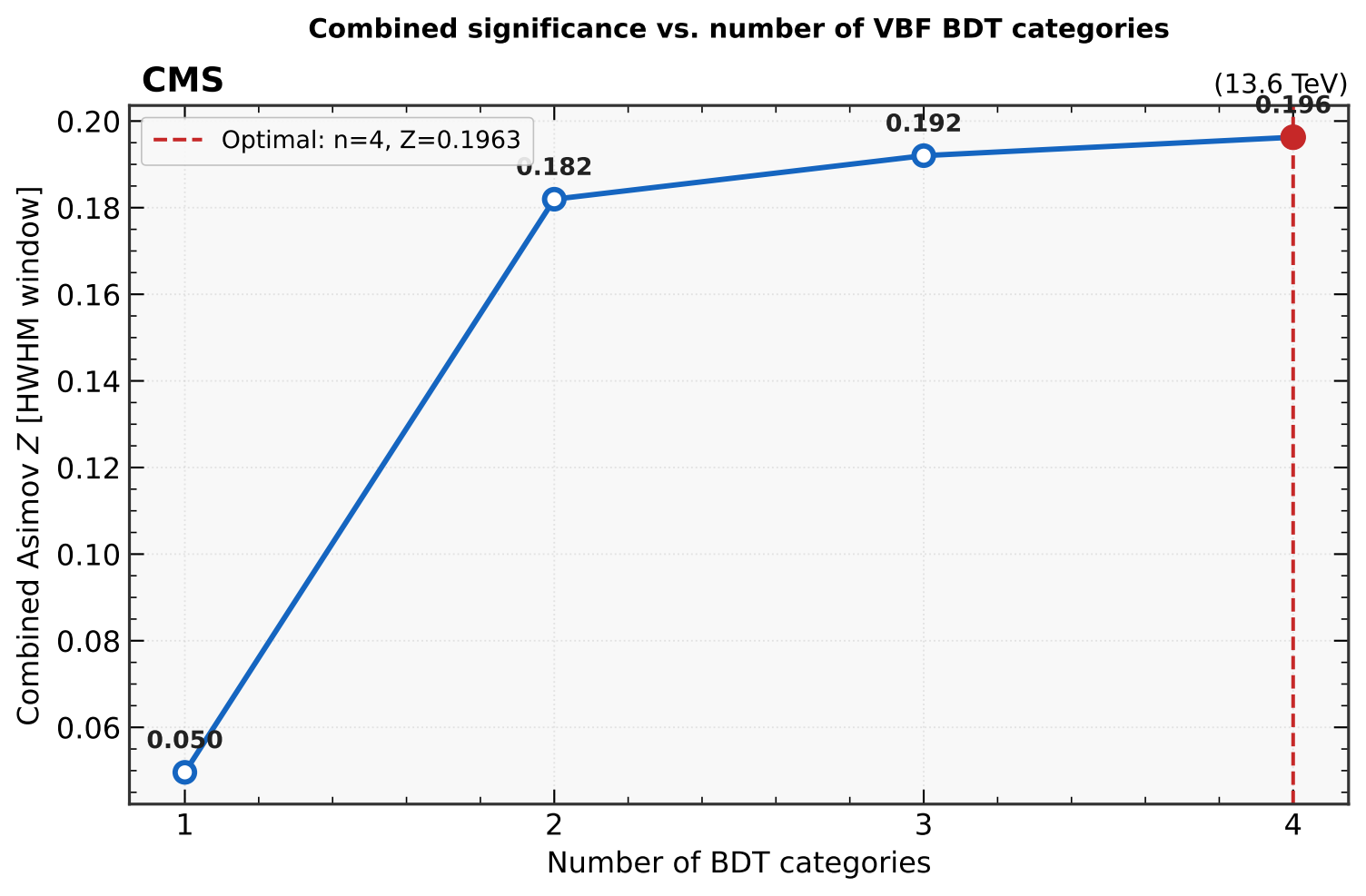

BDT score in full 110–150 GeV window: Data vs. stacked MC
CMS
(13.6 TeV) 22023
Eras: 22023
Full mass window: 110–150 GeV
Boundary range: 0.792 to 0.971
MC total
EWK
DY
Diboson
Top
Data
Events / bin
0
200
400
600
800
Data / Bkg
0.50
0.75
1.00
1.25
1.50
0.0
0.2
0.4
0.6
0.8
1.0
BDT score

BDT bin summary: signal, background, and sensitivity (page 1 / 2)

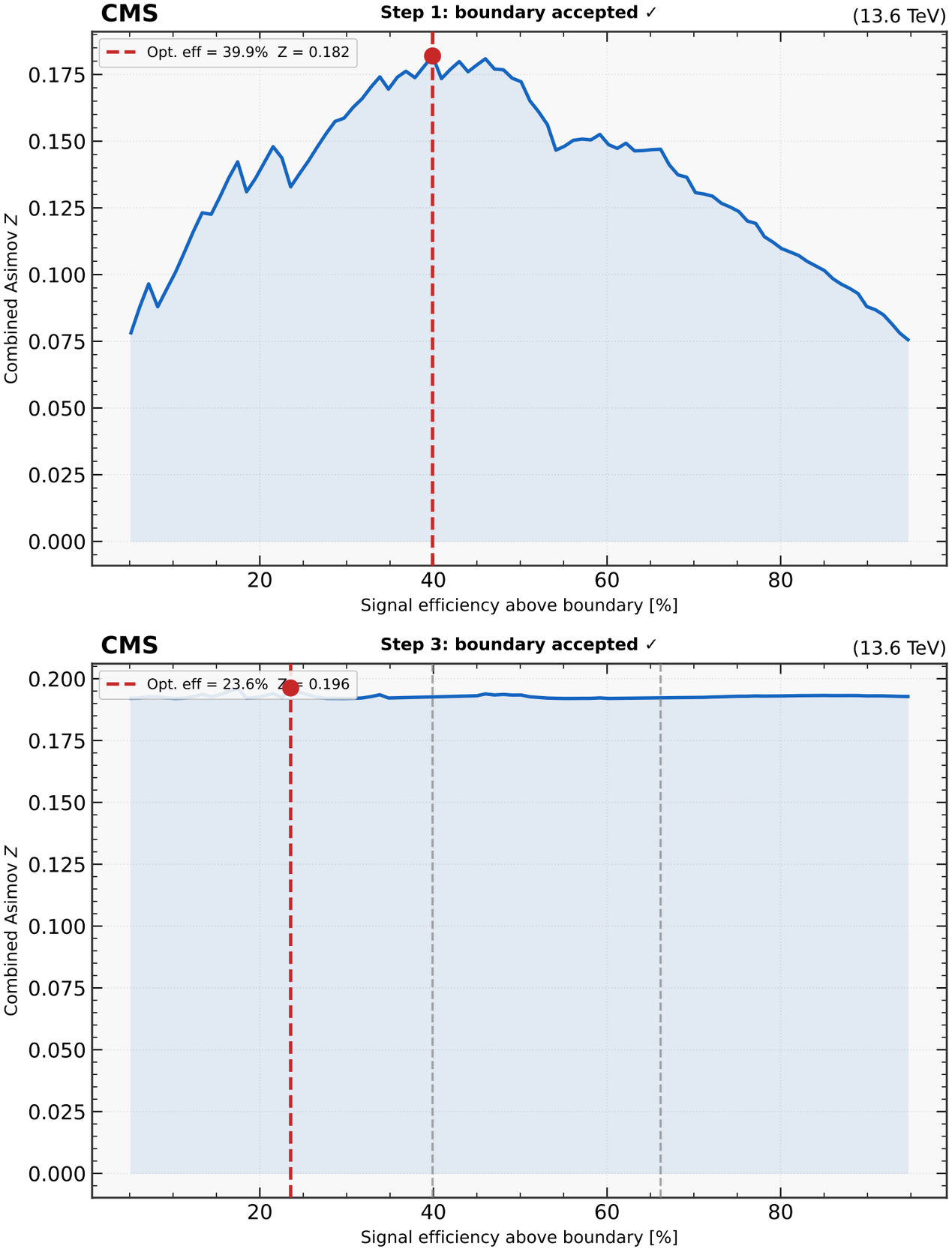
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.000, 0.025)	0.009	918.08 ± 22.01	879	0.00
[0.025, 0.050)	0.016	803.42 ± 20.80	700	0.00
[0.050, 0.075)	0.017	498.71 ± 16.06	585	0.00
[0.075, 0.100)	0.019	409.57 ± 15.00	391	0.00
[0.100, 0.125)	0.019	324.27 ± 13.11	318	0.01
[0.125, 0.150)	0.017	249.53 ± 11.40	215	0.01
[0.150, 0.175)	0.017	229.27 ± 11.25	242	0.01
[0.175, 0.200)	0.018	181.87 ± 9.79	195	0.01
[0.200, 0.225)	0.019	162.53 ± 9.31	169	0.01
[0.225, 0.250)	0.017	158.16 ± 9.46	163	0.01
[0.250, 0.275)	0.019	140.17 ± 8.74	128	0.01
[0.275, 0.300)	0.019	131.91 ± 8.59	141	0.01
[0.300, 0.325)	0.020	108.87 ± 7.69	109	0.02
[0.325, 0.350)	0.020	102.25 ± 7.47	106	0.02
[0.350, 0.375)	0.022	90.66 ± 7.08	101	0.02
[0.375, 0.400)	0.022	95.37 ± 7.29	78	0.02
[0.400, 0.425)	0.023	84.12 ± 6.79	83	0.03
[0.425, 0.450)	0.024	90.38 ± 7.04	78	0.03
[0.450, 0.475)	0.025	75.91 ± 6.50	84	0.03
[0.475, 0.500)	0.026	75.48 ± 6.50	79	0.03

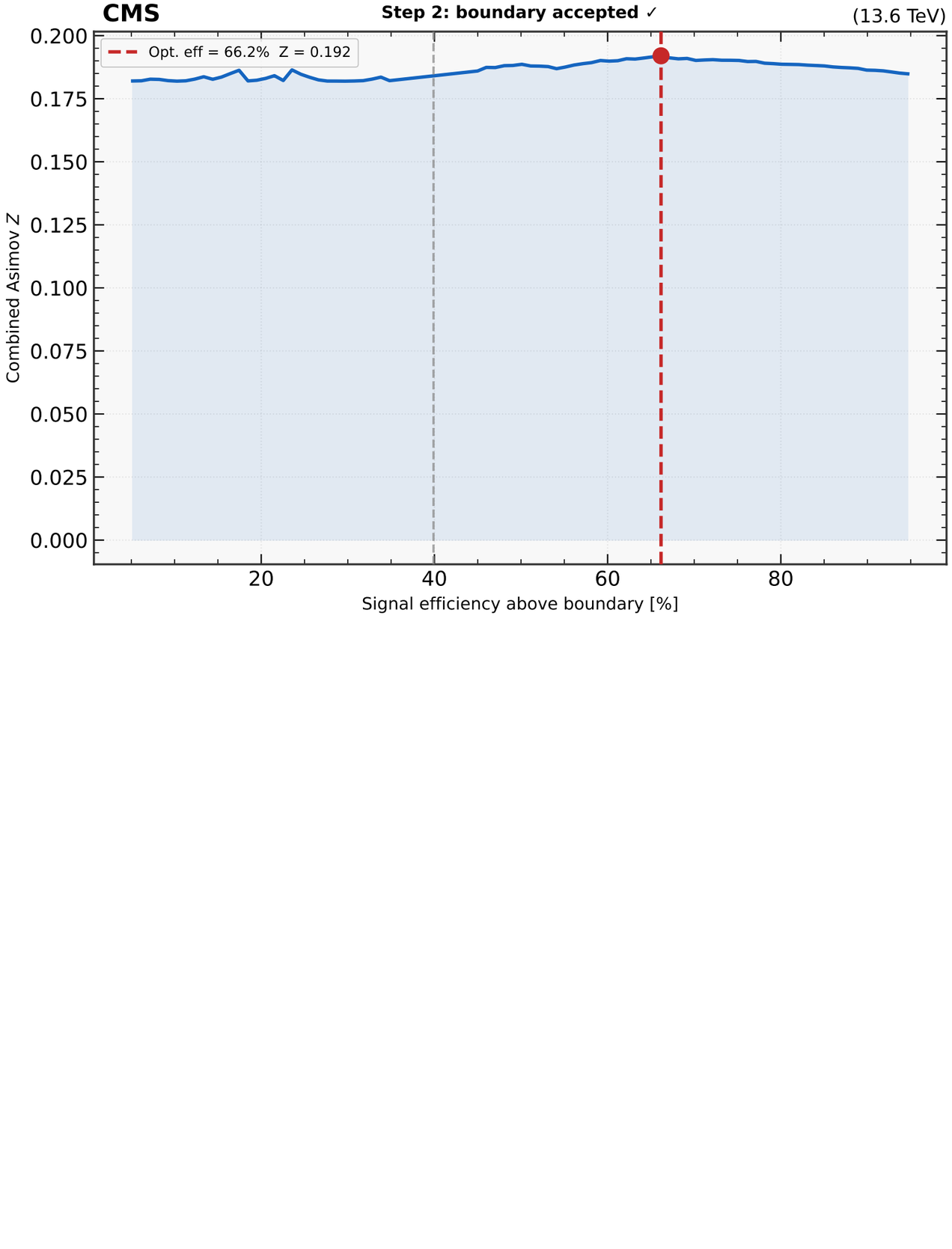
BDT bin summary: signal, background, and sensitivity (page 2 / 2)

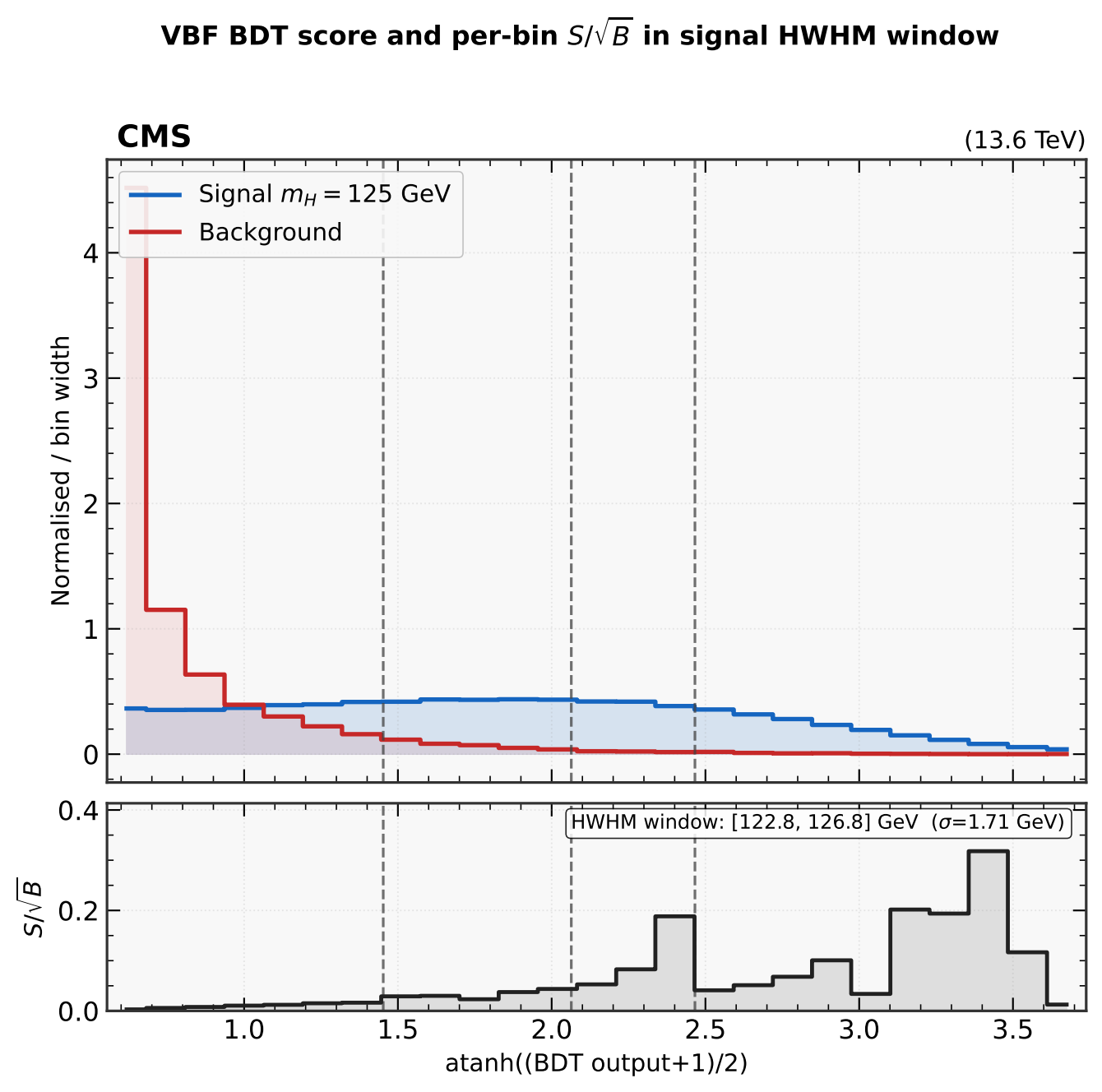
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.500, 0.525)	0.027	65.86 ± 6.01	72	0.04
[0.525, 0.550)	0.029	70.49 ± 6.28	80	0.04
[0.550, 0.575)	0.032	71.02 ± 6.25	52	0.05
[0.575, 0.600)	0.034	59.79 ± 5.73	58	0.06
[0.600, 0.625)	0.035	70.24 ± 6.28	57	0.05
[0.625, 0.650)	0.039	64.01 ± 6.10	59	0.06
[0.650, 0.675)	0.041	59.85 ± 5.77	67	0.07
[0.675, 0.700)	0.045	55.11 ± 5.67	52	0.08
[0.700, 0.725)	0.048	58.82 ± 5.72	60	0.08
[0.725, 0.750)	0.055	54.15 ± 5.60	70	0.10
[0.750, 0.775)	0.060	50.84 ± 5.20	48	0.12
[0.775, 0.800)	0.068	51.27 ± 5.62	40	0.13
[0.800, 0.825)	0.076	47.89 ± 5.02	63	0.16
[0.825, 0.850)	0.089	49.38 ± 5.22	62	0.18
[0.850, 0.875)	0.107	43.04 ± 4.68	53	0.25
[0.875, 0.900)	0.129	44.77 ± 4.99	51	0.29
[0.900, 0.925)	0.165	43.03 ± 4.76	53	0.38
[0.925, 0.950)	0.229	36.61 ± 4.53	64	0.62
[0.950, 0.975)	0.356	41.00 ± 4.90	56	0.86
[0.975, 1.000)	0.549	29.31 ± 4.05	43	1.84

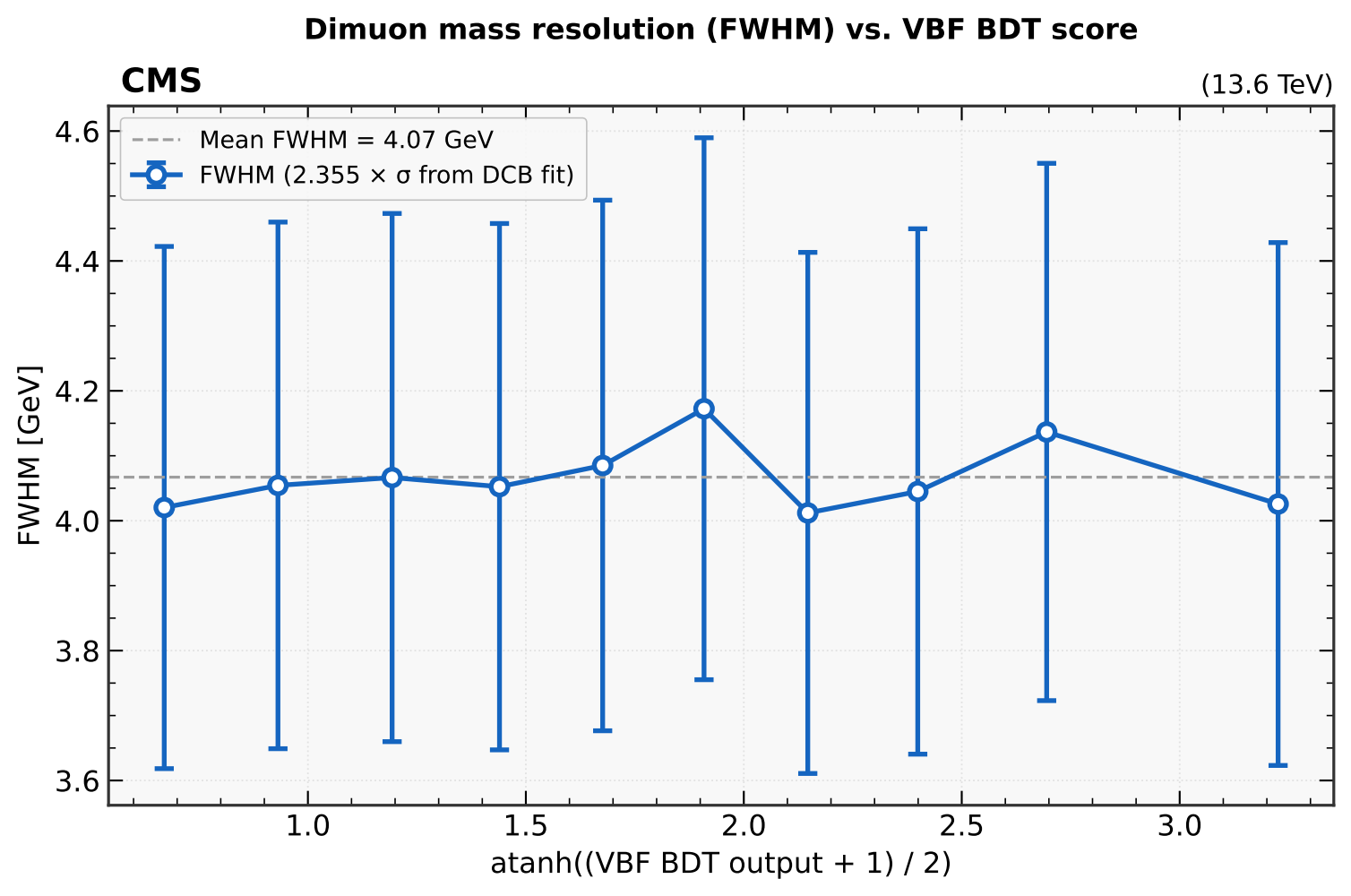

Combined significance vs. number of VBF BDT categories
CMS
(13.6 TeV)
Optimal: n=4, Z=0.1963
0.050
0.182
0.192
0.196
Combined Asimov Z [HWHM window]
Number of BDT categories
0.20
0.18
0.16
0.14
0.12
0.10
0.08
0.06
1
2
3
4

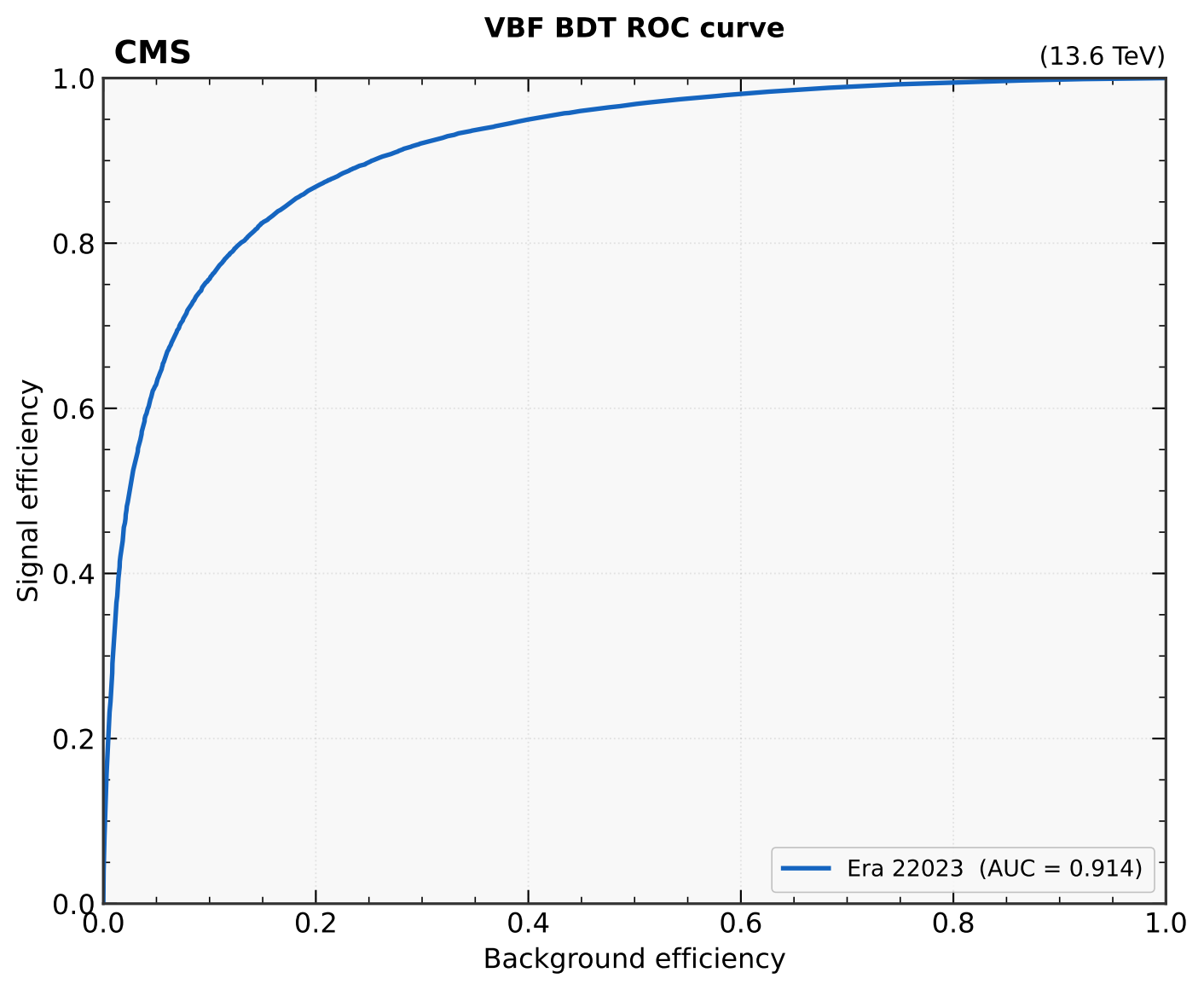
Iterative category boundary optimisation (AN Fig. 32 analog)

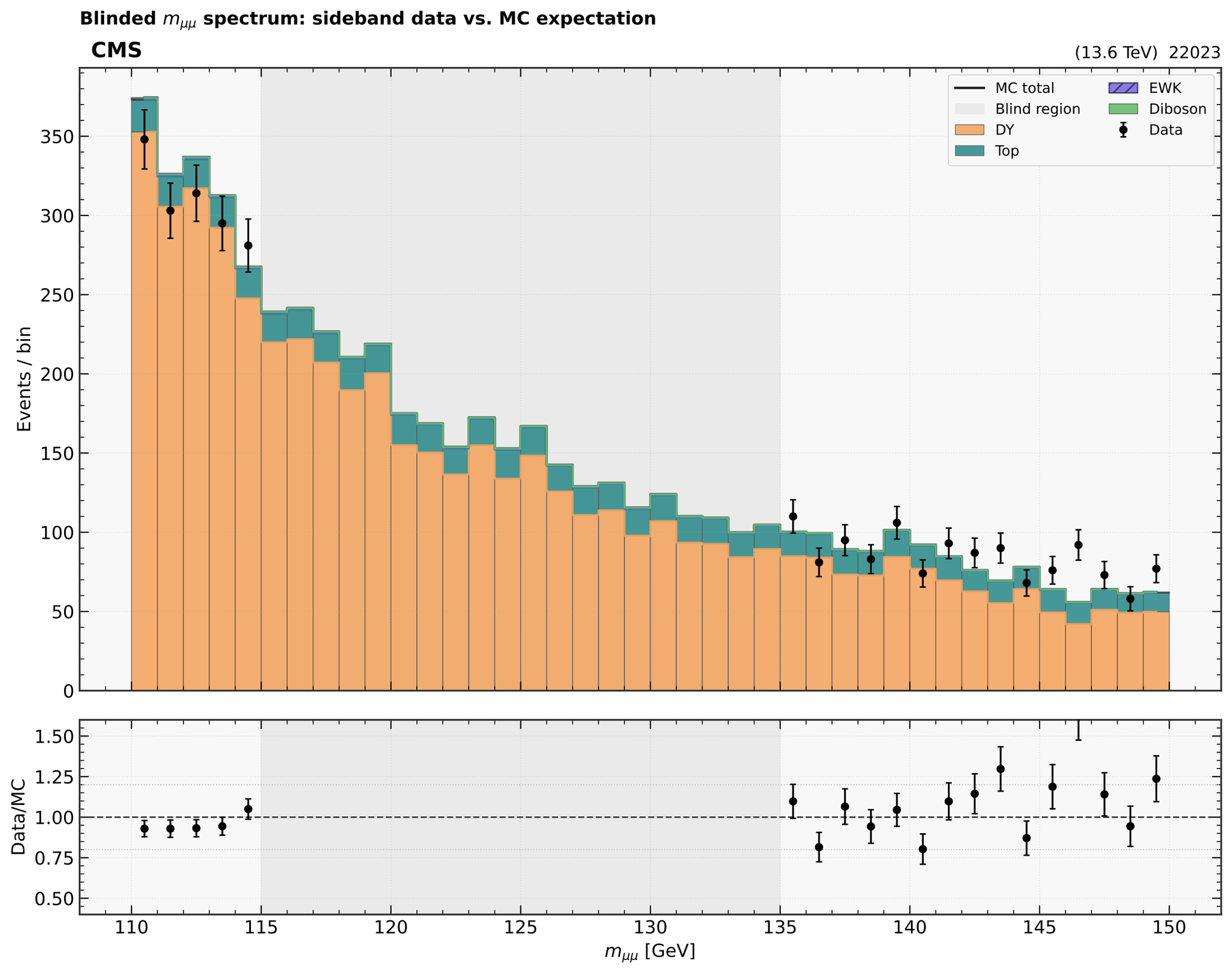


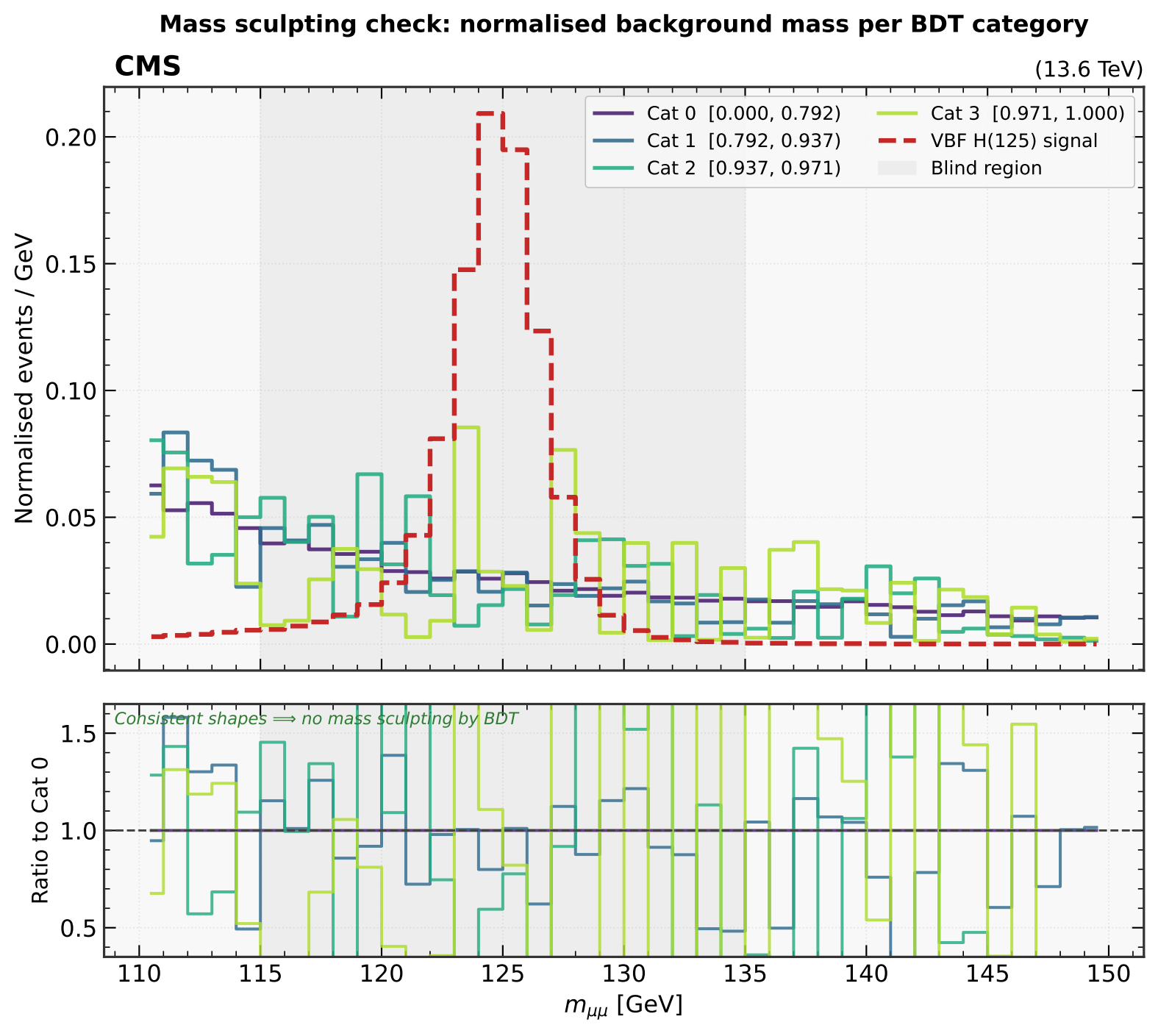



VBF BDT score and per-bin $S/\sqrt{B}$ in signal HWHM window
CMS
(13.6 TeV)
Signal $m_H = 125$ GeV
Background
Normalised / bin width
0
1
2
3
4
HWHM window: [122.8, 126.8] GeV (σ=1.71 GeV)
$S/\sqrt{B}$
0.0
0.2
0.4
1.0
1.5
2.0
2.5
3.0
3.5
atanh((BDT output+1)/2)


Dimuon mass resolution (FWHM) vs. VBF BDT score
CMS
(13.6 TeV)
Mean FWHM = 4.07 GeV
FWHM (2.355 × σ from DCB fit)
FWHM [GeV]
4.6
4.4
4.2
4.0
3.8
3.6
1.0
1.5
2.0
2.5
3.0
atanh((VBF BDT output + 1) / 2)

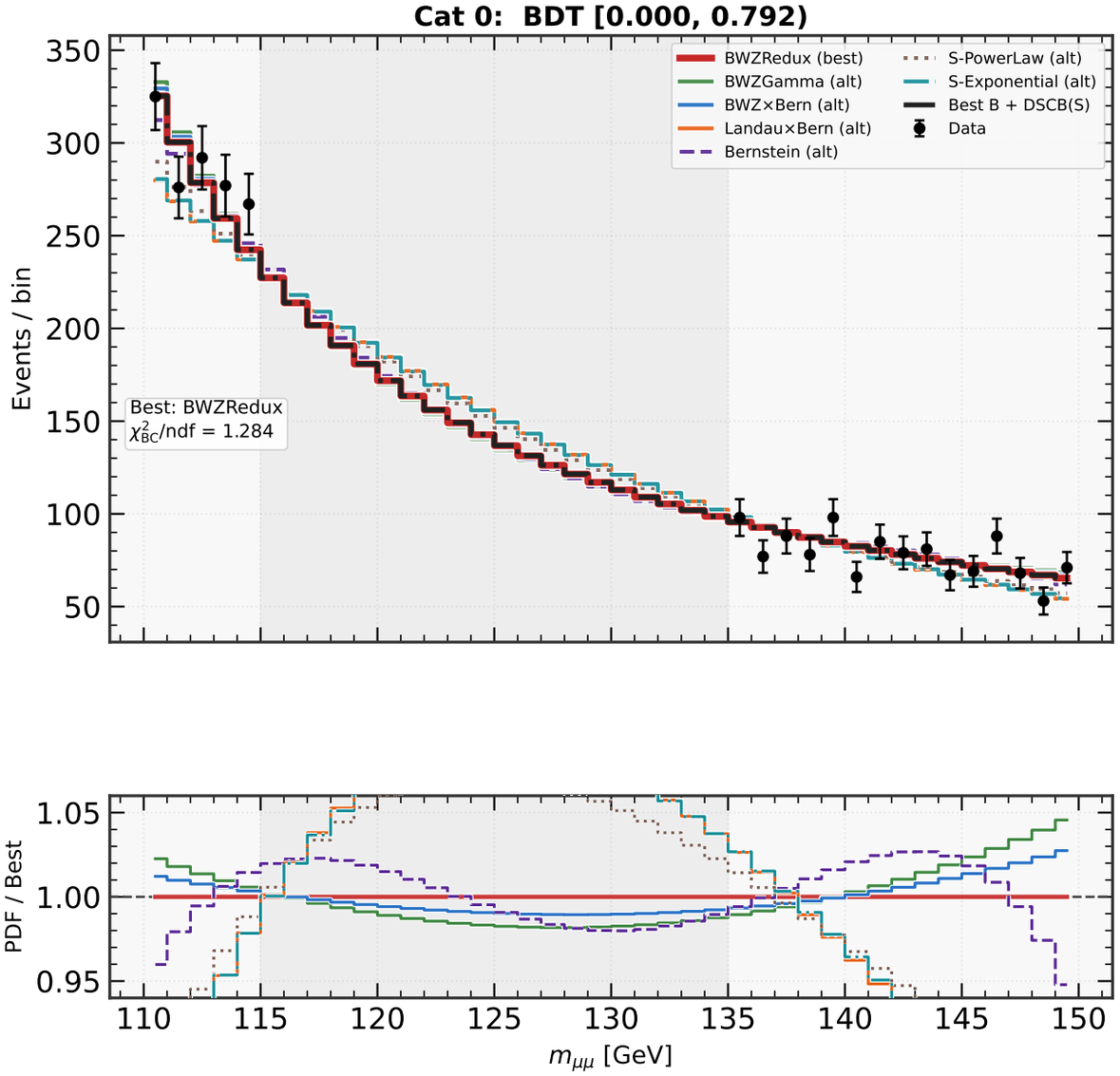

VBF BDT ROC curve
CMS
(13.6 TeV)
Signal efficiency
Background efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Era 22023 (AUC = 0.914)


Blinded $m_{\mu\mu}$ spectrum: sideband data vs. MC expectation
CMS
(13.6 TeV) 2023
MC total
Blind region
DY
Top
EWK
Diboson
Data
Events / bin
Data/MC
$m_{\mu\mu}$ [GeV]

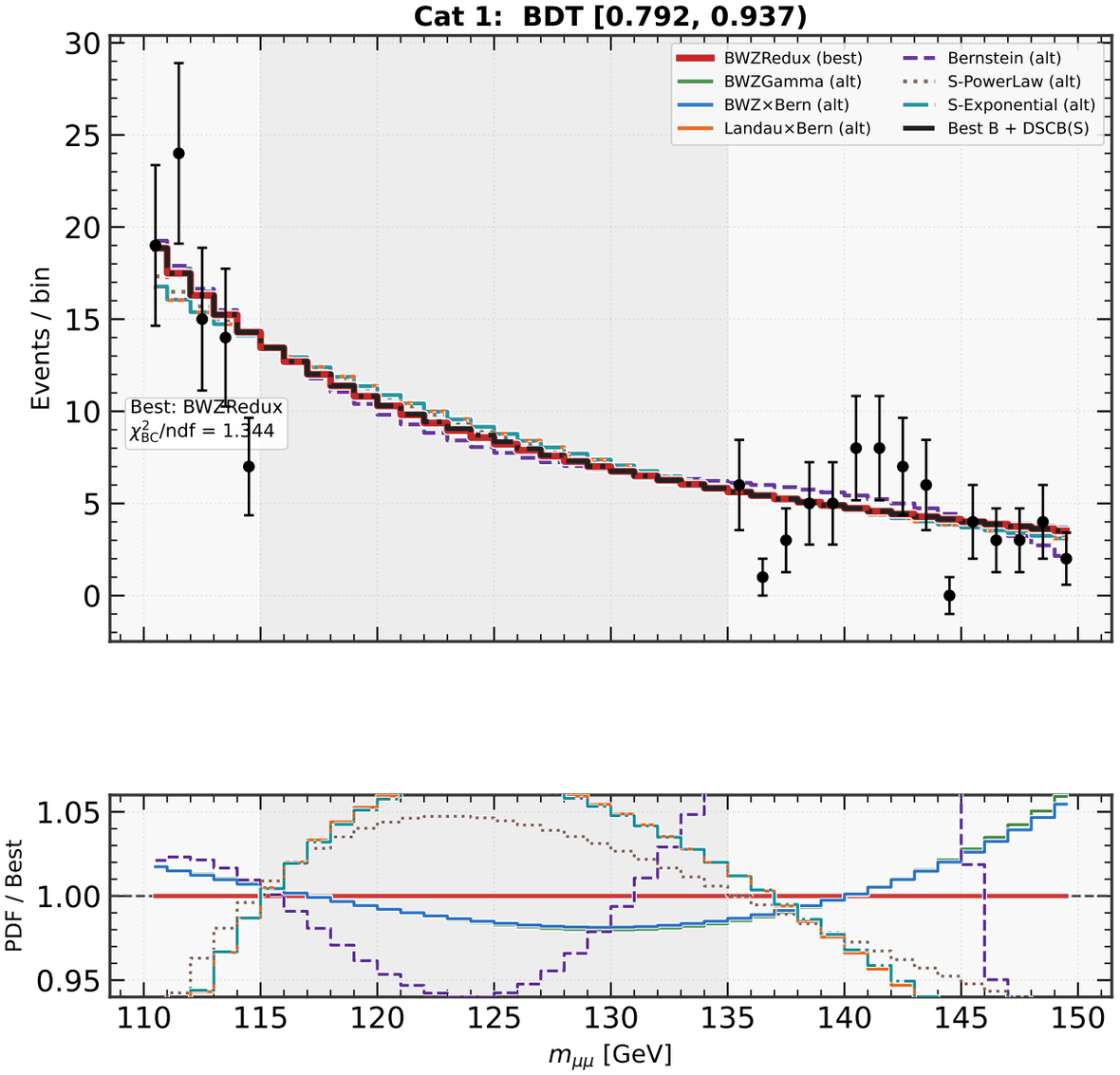

Mass sculpting check: normalised background mass per BDT category
CMS
(13.6 TeV)
Cat 0 [0.000, 0.792)
Cat 1 [0.792, 0.937)
Cat 2 [0.937, 0.971)
Cat 3 [0.971, 1.000)
VBF H(125) signal
Blind region
Normalised events / GeV
0.20
0.15
0.10
0.05
0.00
Consistent shapes ⟹ no mass sculpting by BDT
Ratio to Cat 0
1.5
1.0
0.5
110
115
120
125
130
135
140
145
150
$m_{\mu\mu}$ [GeV]

Per-category dimuon mass: background model fits

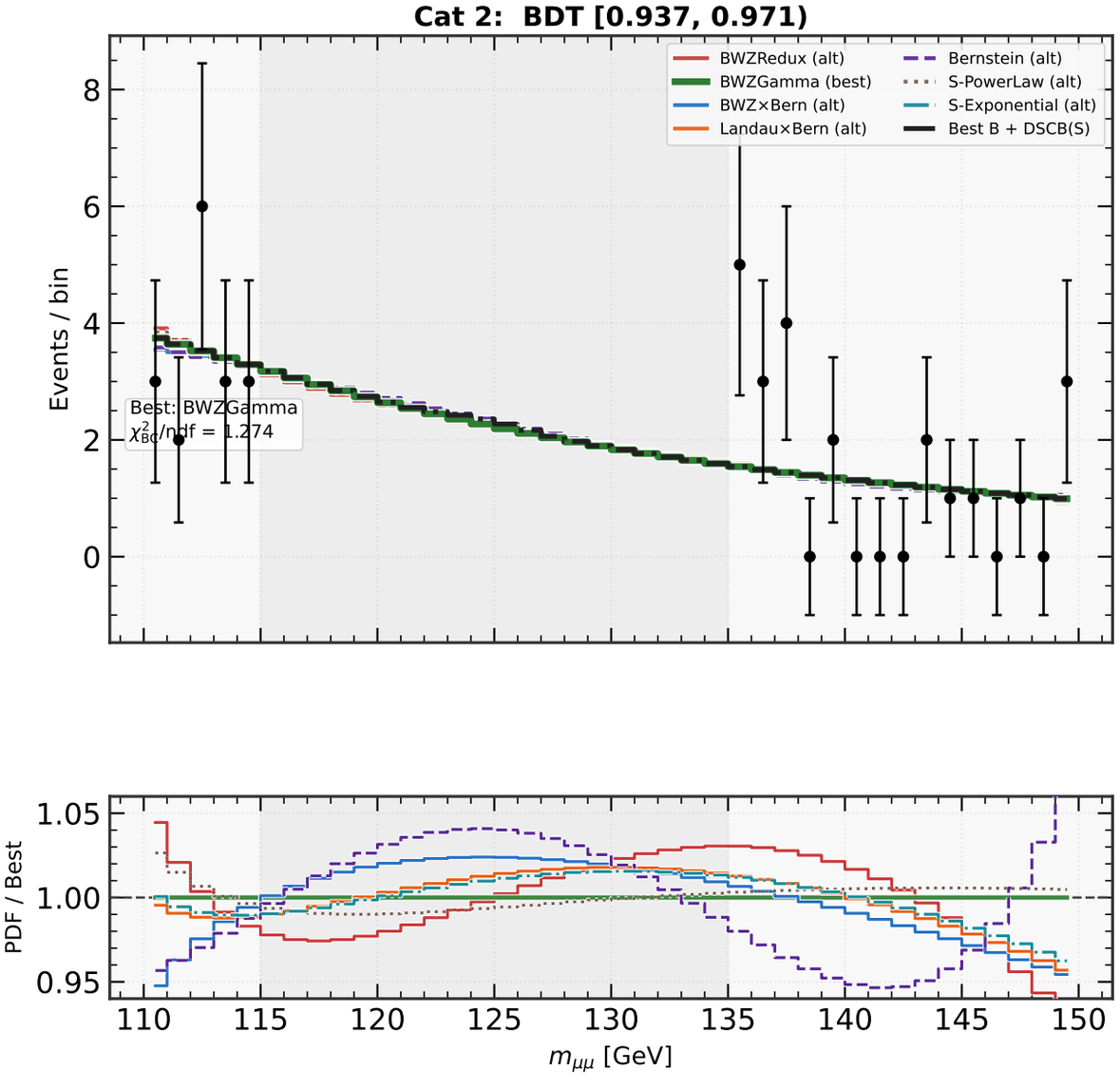
Cat 0: BDT [0.000, 0.792)



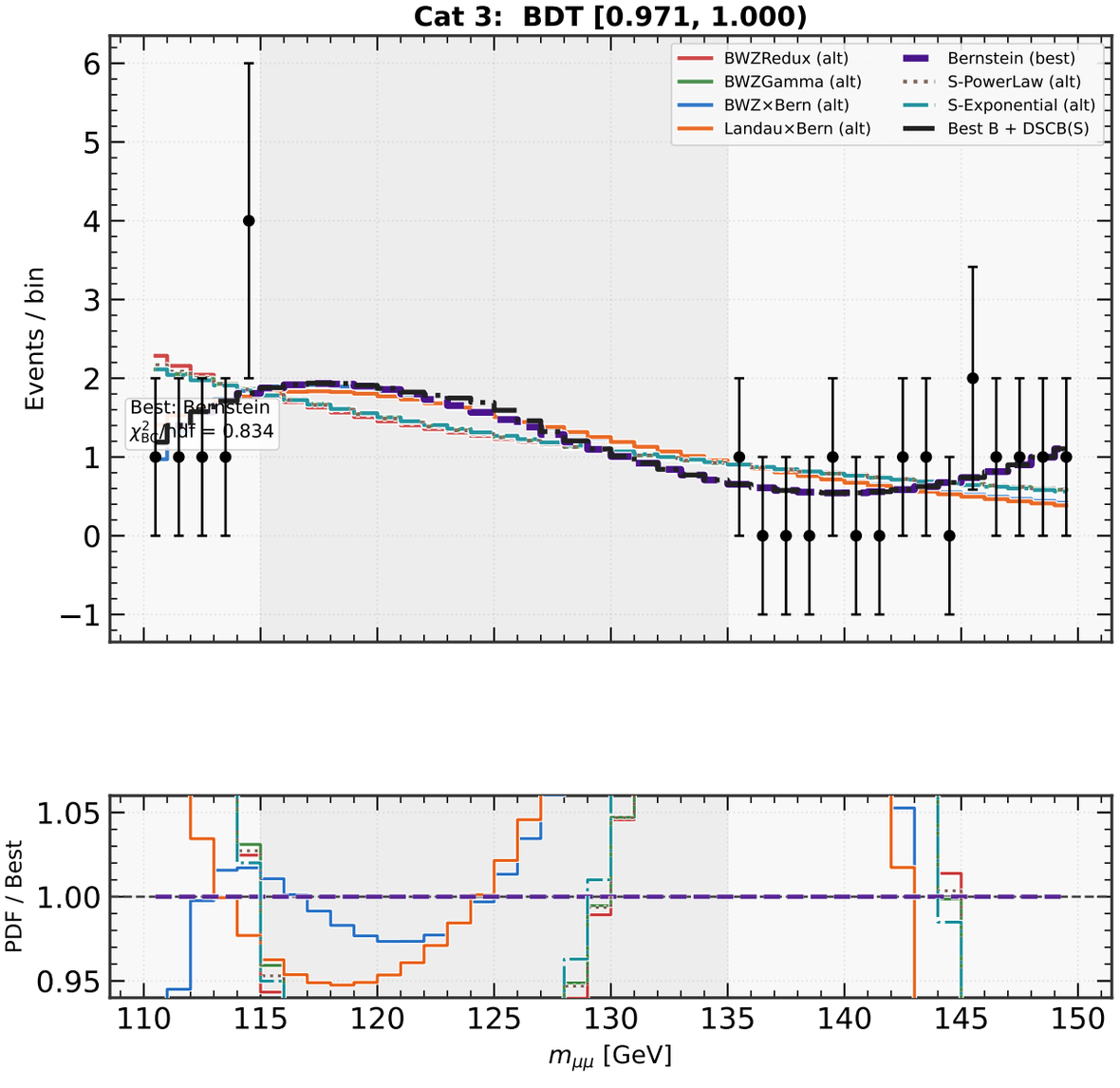
Cat 1: BDT [0.792, 0.937)



Cat 2: BDT [0.937, 0.971)



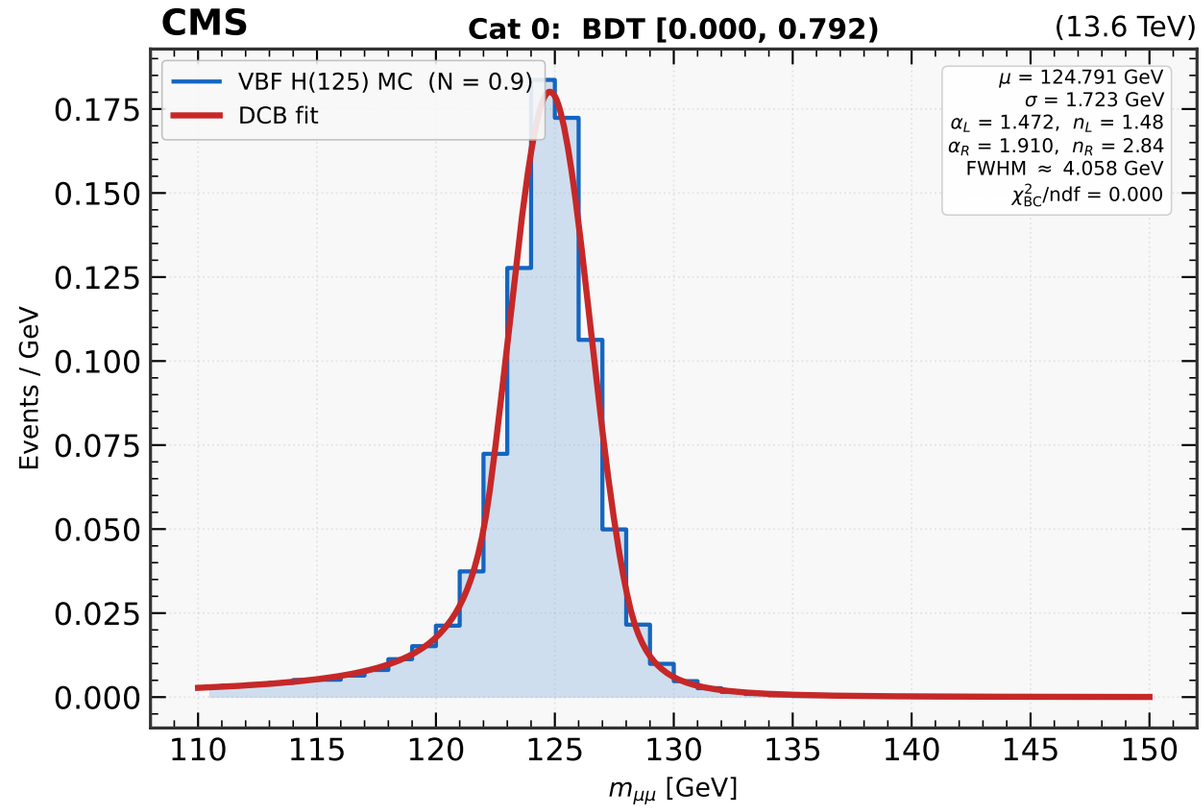
Cat 3: BDT [0.971, 1.000)

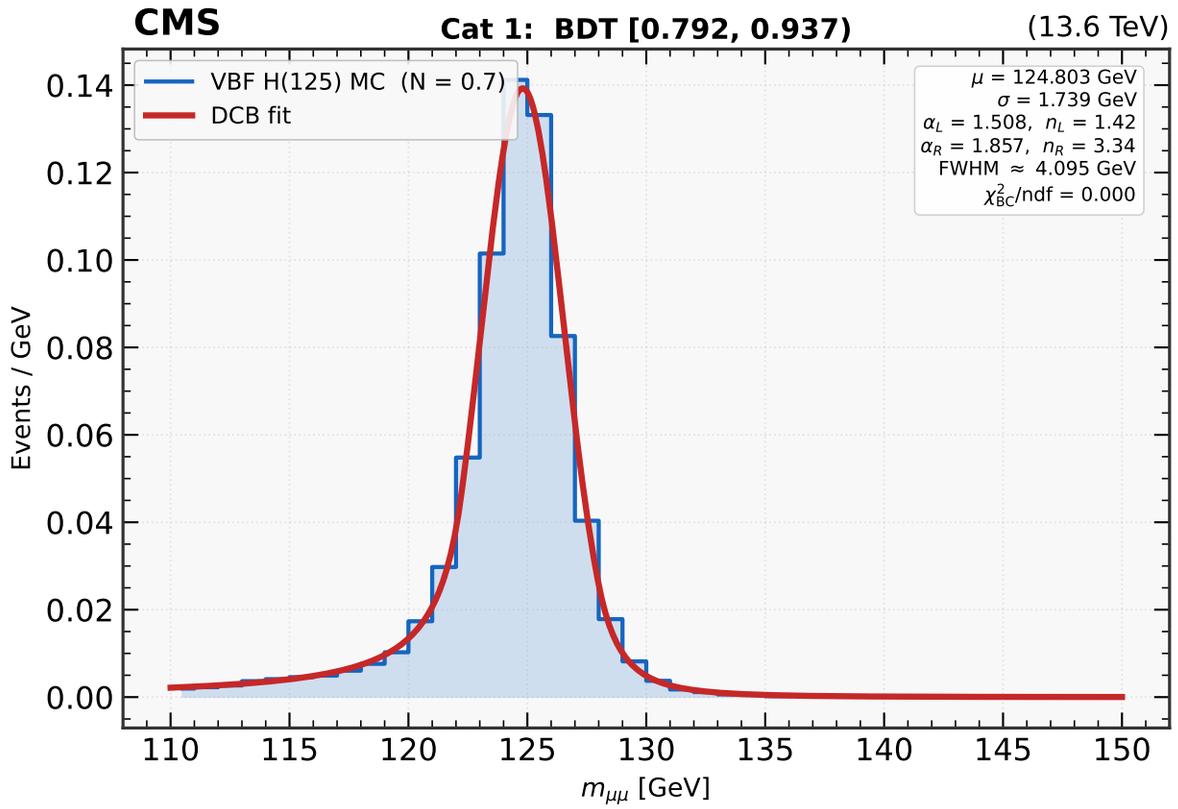


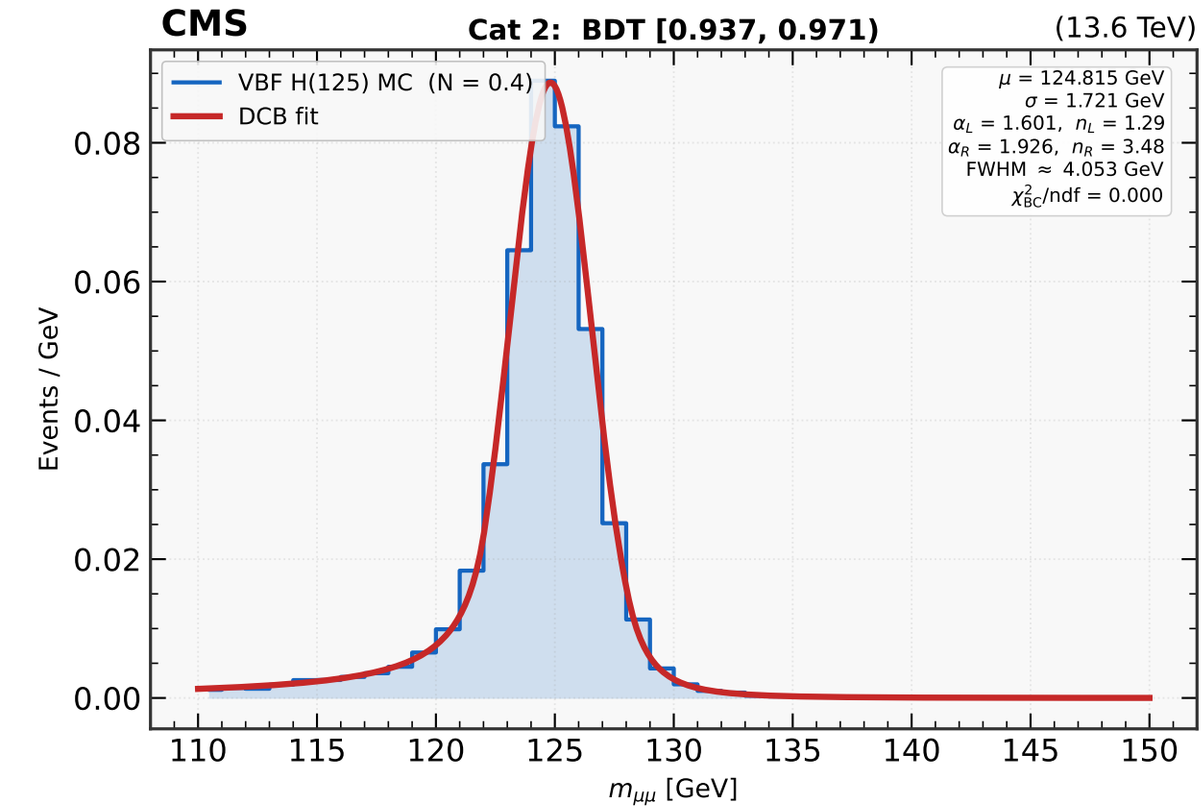
Per-category background model fit comparison (χ²_BC/ndf, ndf)

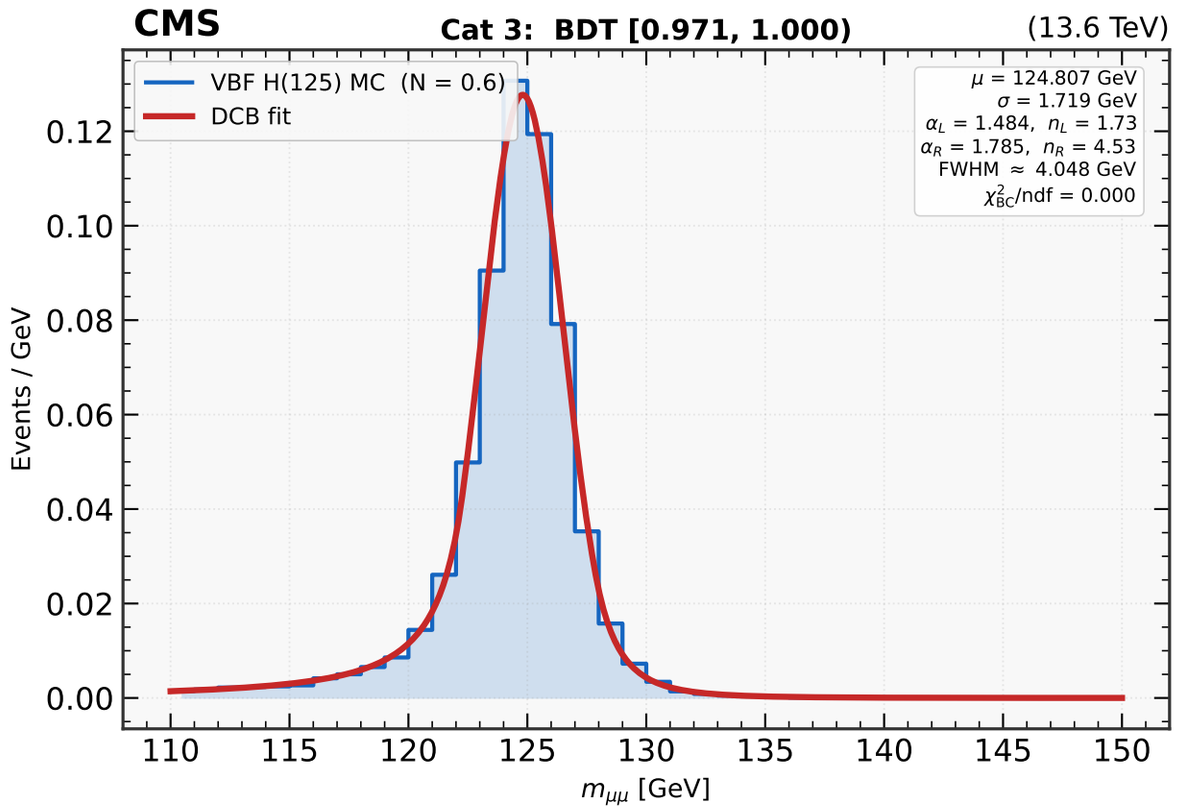
Category	BDT range	Best model	Best χ²_BC/ndf	BWZRedux	BWZGamma	BWZ×Bern	Landau×Bern	Bernstein	S-PowerLaw	S-Exponential
0	[0.000, 0.792)	BWZRedux	1.284	1.284 (37)	1.321 (37)	1.333 (36)	2.278 (37)	1.441 (35)	1.851 (37)	2.235 (37)
1	[0.792, 0.937)	BWZRedux	1.344	1.344 (37)	1.351 (37)	1.387 (36)	1.400 (37)	1.376 (35)	1.381 (37)	1.398 (37)
2	[0.937, 0.971)	BWZGamma	1.274	1.278 (37)	1.274 (37)	1.308 (36)	1.275 (37)	1.345 (35)	1.275 (37)	1.275 (37)
3	[0.971, 1.000)	Bernstein	0.834	0.977 (37)	0.953 (37)	0.937 (36)	0.939 (37)	0.834 (34)	0.959 (37)	0.959 (37)

VBF H(125) signal DCB extraction per BDT category



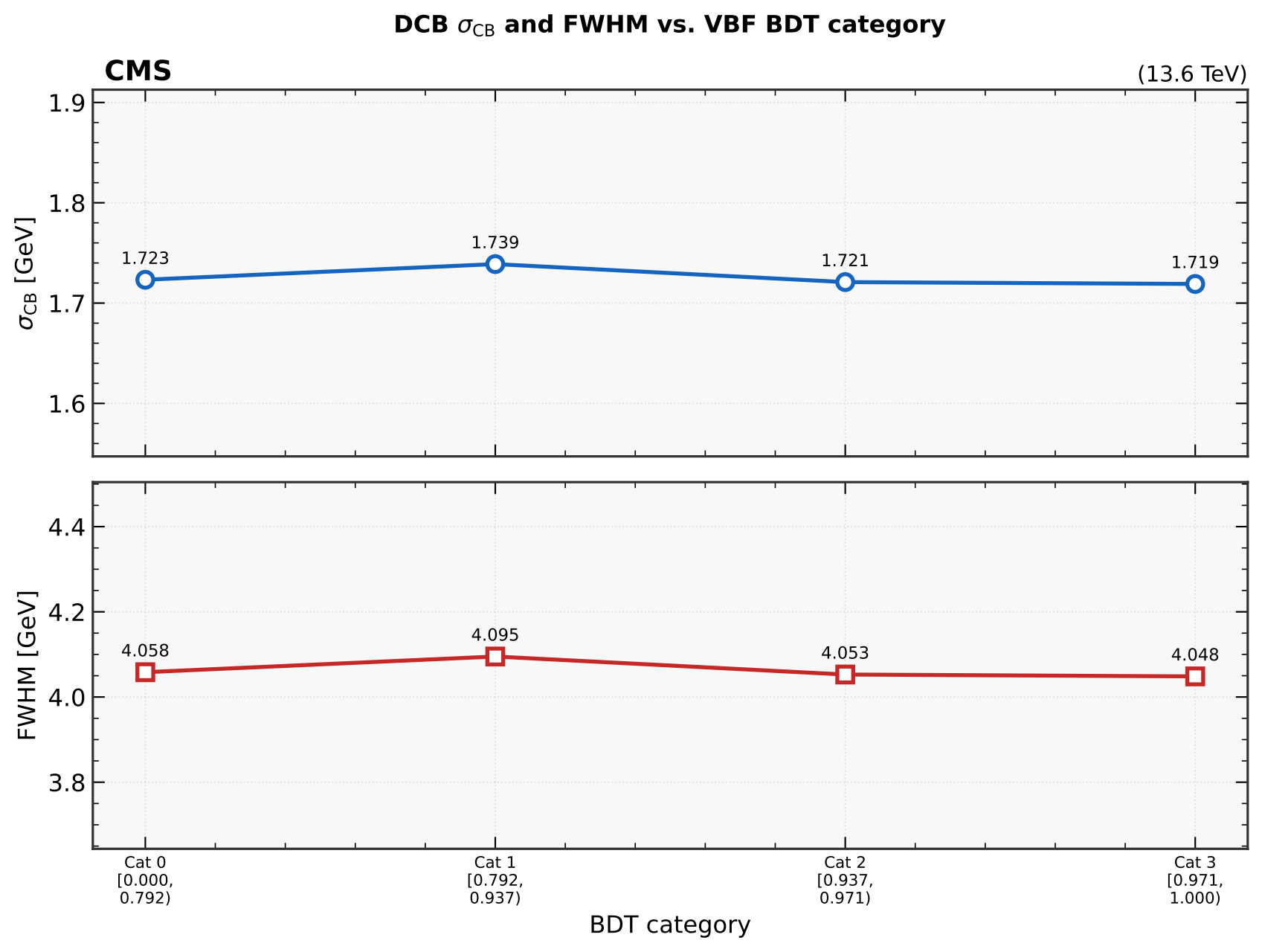


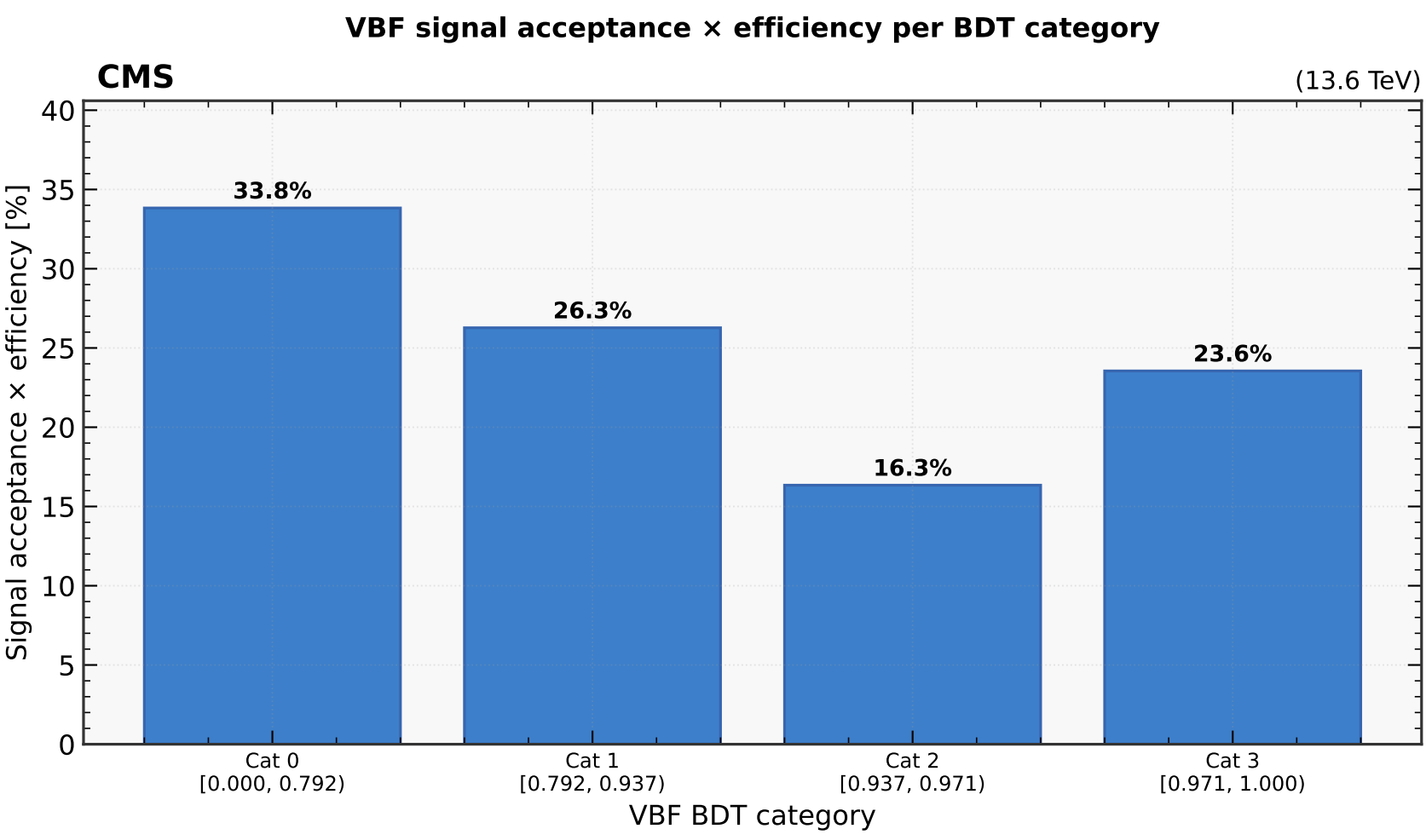




VBF signal DCB fit parameters per category

Category	BDT range	N_sig	Mean [GeV]	Sigma [GeV]	alpha_L	n_L	alpha_R	n_R	FWHM [GeV]	Asimov Z	χ^2_BC/ndf
0	[0.000, 0.792)	0.88	124.7910	1.7233	1.4724	1.482	1.9100	2.842	4.058	0.017	0.000
1	[0.792, 0.937)	0.68	124.8034	1.7389	1.5079	1.425	1.8572	3.343	4.095	0.066	0.000
2	[0.937, 0.971)	0.43	124.8148	1.7209	1.6012	1.292	1.9265	3.482	4.053	0.126	0.000
3	[0.971, 1.000)	0.61	124.8067	1.7191	1.4840	1.728	1.7851	4.526	4.048	0.128	0.000


DCB σ_{CB} and FWHM vs. VBF BDT category
CMS
(13.6 TeV)
σ_{CB} [GeV]
1.9
1.8
1.7
1.6
1.723
1.739
1.721
1.719
FWHM [GeV]
4.4
4.2
4.0
3.8
4.058
4.095
4.053
4.048
Cat 0 [0.000, 0.792)
Cat 1 [0.792, 0.937)
Cat 2 [0.937, 0.971)
Cat 3 [0.971, 1.000)
BDT category


VBF signal acceptance × efficiency per BDT category
CMS
(13.6 TeV)
Signal acceptance × efficiency [%]
0
5
10
15
20
25
30
35
40
33.8%
26.3%
16.3%
23.6%
Cat 0
[0.000, 0.792)
Cat 1
[0.792, 0.937)
Cat 2
[0.937, 0.971)
Cat 3
[0.971, 1.000)
VBF BDT category

Per-category yield and sensitivity summary

Category	BDT range	S_full	B_full	S_HWHM	B_HWHM	HWHM window [GeV]	S/√B	Asimov Z
0	[0.000, 0.792)	0.88	11298.61	0.599	1217.964	[122.8, 126.8]	0.017	0.017
1	[0.792, 0.937)	0.68	521.53	0.467	50.171	[122.8, 126.9]	0.066	0.066
2	[0.937, 0.971)	0.43	100.14	0.292	5.246	[122.8, 126.8]	0.128	0.126
3	[0.971, 1.000)	0.61	76.17	0.424	10.859	[122.8, 126.8]	0.129	0.128

Category summary: HWHM-windowed signal and background yields

Category	BDT range	Data sidebands	MC sidebands ± stat	Data/MC (sidebands)	MC full (110-150) ± stat	S/√B [HWHM]	Asimov Z (cat) [HWHM]	B_HWHM
0	[0.000, 0.792)	2603	5276.91 ± 75.64	0.493	11298.61 ± 110.48	0.0172	0.0172	1217.964
1	[0.792, 0.937)	144	252.83 ± 16.52	0.570	521.53 ± 23.91	0.0660	0.0659	50.171
2	[0.937, 0.971)	39	42.31 ± 7.09	0.922	100.14 ± 10.63	0.1276	0.1264	5.246
3	[0.971, 1.000)	18	37.08 ± 6.43	0.485	76.17 ± 9.46	0.1286	0.1278	10.859
Combined	all						0.1922	

Category composition by process (full mass window)

Category	BDT range	DY	EWK	Top	Diboson	total_err2	total
0	[0.000, 0.792)	10019.60	15.52	1241.29	22.20	12206.65	11298.61
1	[0.792, 0.937)	464.88	9.41	45.23	2.01	571.92	521.53
2	[0.937, 0.971)	86.65	5.12	7.81	0.55	112.98	100.14
3	[0.971, 1.000)	65.71	6.19	4.06	0.21	89.58	76.17